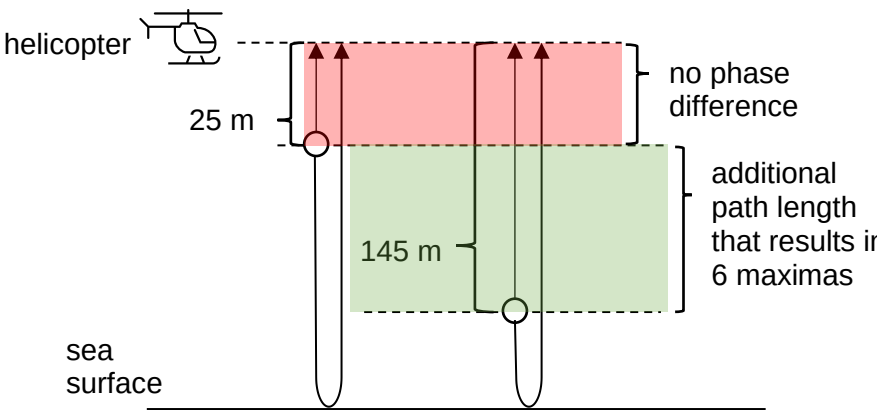


Qns	Answer	Marks
8(a)(i)	when two or more waves meet and overlap, resultant displacement is vector sum of displacement of each individual wave	1
8(a)(ii)1.	waves from emitter directly and after reflecting off the sea surface meet and overlap at receiver/helicopter different path lengths as emitter drops so waves arrive at receiver/helicopter changing phase difference maxima when waves arrive in phase and undergo constructive interference minima when waves arrives in anti-phase and undergo destructive interference	1 1 1
8(a)(ii)2.	(words to effect of) attenuation of waves increase with distance from emitter / waves have lower intensity at greater distance from emitter / are of lower amplitude at greater distance from emitter (words to effect of) when emitter is near helicopter, amplitude/intensity of reflected wave is lower OR reflected wave is more attenuated than wave reaching directly so incomplete destructive interference OR (words to effect of) when emitter is near water surface, path length of both waves reaching helicopter directly and reflected wave is similar. so waves reach helicopter with similar intensity / amplitude / attenuation, resulting in more complete destructive interference	1 1
8(a)(iii)	(words to effect of) 6 maximas between $d = 25$ m and $d = 145$ m (words to effect of) distance between adjacent maximas is half wavelength $6\left(\frac{\lambda}{2}\right) = 145 - 25$ $\lambda = 40 \text{ m}$ 	1 1 1
8(b)(i)	splitting of a single heavy nucleus when bombarded by neutrons to form two or more lighter nuclei of approximately same mass with neutrons emitted	1
8(b)(ii)	spontaneous and random emission of ionizing radiation in the form of alpha particles, beta particles or gamma ray photons from unstable nucleus to become a more stable nucleus	1

Qns	Answer	Marks
8(b)(iii)	combining of two or more light nuclei under very high temperatures to form a single, more massive nucleus	1
8(c)(i)	allows values of different orders of magnitude to be shown on one graph / along one axis	1
8(c)(ii)	[answer deals with conservation of mass in the form of nucleon number] fission results in two daughter nuclei and two neutrons of <u>total nucleon number 236</u> ${}_{92}^{235}\text{U} + {}_0^1\text{n} \rightarrow {}_{92-y}^{117+x}\text{A} + {}_y^{117-x}\text{B} + 2{}_0^1\text{n}$ if same mass, each product has nucleon number of 117 $\left(= \frac{1}{2}(236 - 2) \right)$ if one fission product has a nucleon number less than 117, the other fission product have the same difference in nucleon number above 117 therefore symmetrical about 117	1 1 1 A0
8(d)(i)	${}_{53}^{140}\text{I} \rightarrow {}_{53}^{139}\text{I} + {}_0^1\text{n}$	1
8(d)(ii)	${}_{53}^{140}\text{I} \rightarrow {}_{54}^{140}\text{Xe} + {}_{-1}^0\text{e} \quad \left(+\bar{\nu} \right)$	1
8(d)(i)	energy released = [(total mass of reactant) – (total mass of product)] uc^2 $= [139.9019 - (138.8969 + 1.0087)]uc^2$ $= -0.0037uc^2 \quad \text{OR} \quad -5.58 \times 10^{-13} \text{ J} \quad \text{OR} \quad -3.49 \times 10^6 \text{ eV}$ not feasible	1
8(d)(i)	energy released = [(total mass of reactant) – (total mass of product)] uc^2 $= [139.9019 - (139.8919 + 0.0006)]uc^2$ $= 0.0094uc^2 \quad \text{OR} \quad 1.40 \times 10^{-12} \text{ J} \quad \text{OR} \quad 8.75 \times 10^6 \text{ eV}$ feasible	1

Qns	Answer	Marks
Q(Vi)	[source of force] wire experiences magnetic force	1